

Equal Gaussian cells: the sharp simplex first-moment theorem

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Abstract

For every number of cells $k \geq 2$, we determine the sharp squared first-moment sum for equal-probability Gaussian partitions when the ambient dimension is at least $k-1$. The unique maximizers are central regular simplex cones, and the value is $(\mathbb{E} \max_{i \leq k} Z_i)^2 / (k-1)$ for independent standard normals. The bound also holds in lower dimensions and for fractional partitions. We prove a covariance comparison by combining Gaussian flux with the established Gaussian multi-bubble perimeter theorem. A weighted Cauchy inequality yields a first-order differential inequality along each path from the regular covariance; its initial derivative vanishes by symmetry. This gives the global bound and equality classification, with an exact nonnegative integral expression for the deficit. No numerical search, variation of arbitrary competing cell boundaries, local simplex Hessian or mountain-pass argument is needed. We distinguish the imported perimeter theorem and partial formal checks from the new analytic comparison.

1. Statement and prior conjecture

Let γ_d be standard Gaussian probability on $\mathbb{R}^d$. For a measurable partition $C_1, \dots, C_k$ define $m_i = \int_{C_i} x d\gamma_d$ and $F = \sum_i |m_i|^2$. Write $Z_1, \dots, Z_k$ for independent standard normals and

$$a_k = \mathbb{E} \max_{i \leq k} Z_i, \quad c_k = \frac{a_k}{\sqrt{k-1}}, \quad F_k = c_k^2.$$

These constants are exact; for example integration by parts gives

$$a_k = k(k-1) \int_{\mathbb{R}} \phi(x)^2 \Phi(x)^{k-2} dx > 0, \tag{1}$$

where ϕ, Φ are the standard normal density and distribution.

Theorem 1. For every integer $k \geq 2$, every integer $d \geq 1$, and every Gaussian partition with $\gamma_d(C_i) = 1/k$,

$$\sum_{i=1}^k \left| \int_{C_i} x d\gamma_d(x) \right|^2 \leq \frac{a_k^2}{k-1}. \tag{2}$$

If $d \geq k-1$, equality holds precisely for the central regular simplex winning partition in a $(k-1)$ -dimensional subspace, with the orthogonal complement unrestricted, up to null sets, orthogonal

transformations and relabeling. If $d < k - 1$, the inequality is strict for every partition; we do not claim this is the sharp bound at a fixed such lower dimension.

The identical statement holds for fractional partitions: $0 \leq f_i \leq 1$, $\sum_i f_i = 1$ almost everywhere, and $\int f_i d\gamma_d = 1/k$, using $m_i = \int x f_i d\gamma_d$. The functions f_i are measurable. Equality forces the same ordinary winning indicators.

This settles the equal-mass subcase for every number of cells of the first-moment conjecture in [H19, Problem 1.15 and Conjecture 1.16], and in particular the four-cell dimension-three question [H14, Conjecture 3]. It is not the positive-correlation noise-stability conjecture or the arbitrary prescribed-mass statement. The latter is false as written [EX]. Our literature status reports the precise sources checked and makes no worldwide first-proof claim.

2. The balanced covariance value function

Put $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^k$, $P = I - \mathbf{1}\mathbf{1}^T/k$, $n = k - 1$,

$$\mathcal{K} = \{Q = Q^T \succeq 0 : Q\mathbf{1} = 0, \text{tr } Q = 1\}, \quad Q_* = P/n.$$

For every positive semidefinite Q annihilating $\mathbf{1}$, define

$$\mathcal{C}(Q) = \inf_{\lambda \in \mathbb{R}^k} \left\{ \frac{1}{k} \sum_i \lambda_i + \mathbb{E} \max_i (Y_i - \lambda_i) \right\}, \quad Y \sim N(0, Q). \quad (3)$$

For a centered $k \times n$ score matrix M with $MM^T = Q$, this is called the balanced score value. For distinct rows it equals the optimal assignment value, as proved below. Prices attain their infimum: in the gauge $\min \lambda_i = 0$, the expression is at least $\max \lambda_i/k$ by using a zero-priced mean-zero score. This closed gauge therefore has bounded sublevel sets, and the objective is continuous. The maximum also bounds the average, so the objective is nonnegative. A compactness argument gives a minimum; common price shifts let us place it in the smooth gauge $\sum_i \lambda_i = 0$. For distinct scores v_i , differentiating in prices gives $1/k - \gamma_n(C_i)$, where

$$C_i = \{x : v_i \cdot x - \lambda_i \geq v_j \cdot x - \lambda_j \ \forall j\}. \quad (4)$$

Ties are Gaussian-null. Minimizing prices therefore give all masses $1/k$; the winning labels attain the linear assignment pointwise.

Two elementary facts include the singular covariance boundary:

$$\mathcal{C}(sQ) = \sqrt{s} \mathcal{C}(Q) \ (s \geq 0), \quad \mathcal{C} \text{ is continuous on the closed cone.} \quad (5)$$

For homogeneity scale scores and prices, and note $\mathcal{C}(0) = 0$. For continuity couple $Q^{1/2}G$ and $\tilde{Q}^{1/2}G$ on the same standard Gaussian space. The difference of the maxima, for any price, is at most $|(Q^{1/2} - \tilde{Q}^{1/2})G|$. Its expectation bounds the difference of the two infima, uniformly in prices; the matrix square root is continuous.

We use differentiability only where Q is positive definite on $\mathbf{1}^\perp$. There the k scores are affinely independent in $\mathbb{R}^n$. Every pair of winning cells has a positive Gaussian facet area: the nonsingular

score-difference coordinates allow any chosen pair to tie at the top while all other scores are strictly lower. For the ordinary ij facet put

$$\ell_{ij} = |v_i - v_j|, \quad A_{ij} = \int_{\Sigma_{ij}} \varphi_n d\mathcal{H}^{n-1}, \quad w_{ij} = A_{ij}/\ell_{ij}, \quad (Lu)_i = \sum_{j \neq i} w_{ij}(u_i - u_j). \quad (6)$$

Let B have rows $b_i = \int_{C_i} x d\gamma_n$. Gaussian integration by parts gives

$$B = LM, \quad \mathcal{C}(Q) = \langle M, B \rangle = \text{tr}(LQ) = \sum_{i < j} w_{ij} \ell_{ij}^2, \quad \text{tr } L = 2 \sum_{i < j} w_{ij}. \quad (7)$$

The inward ij normal of cell i is $(v_i - v_j)/\ell_{ij}$. To justify flux on unbounded cells truncate with large balls; their additional Gaussian boundary term tends to zero exponentially. The price term in (3) cancels because all winning masses are $1/k$.

Lemma 2 (smooth derivative). On this positive cone $\mathcal{C}$ is smooth, and for every symmetric D annihilating $\mathbf{1}$,

$$d\mathcal{C}(Q)[D] = \frac{1}{2} \text{tr}(LD). \quad (8)$$

Proof. Let T have rows $v_i - v_k$, $i < k$, and put $a_i = \lambda_i - \lambda_k$. The map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible. With $s = TX - a$, its Gaussian covariance is $TT^T > 0$ and its mean is $-a$. If $p_{\Sigma, \mu}$ denotes that ordinary n -dimensional Gaussian density, the objective equals

$$J(M, \lambda) = \frac{1}{k} \sum_i \lambda_i - \lambda_k + \int_{\mathbb{R}^n} \max(0, s_1, \dots, s_n) p_{TT^T, -a}(s) ds. \quad (8a)$$

The winner regions in s are fixed polyhedral cones. On a compact neighborhood of a full-rank score and finite price, covariance eigenvalues stay bounded away from zero and infinity and the mean stays bounded. Each parameter derivative of the density is bounded by $C(1 + |s|)^r e^{-c|s|^2}$ for constants $C, c > 0$ and a finite r depending on the derivative order. The maximum has at most linear growth. Differentiation under the integral therefore proves smoothness of the objective and all winning probabilities.

For a fixed cell i , use the invertible coordinates $z_j = (v_i - v_j) \cdot X$, $j \neq i$. Its probability is a Gaussian orthant integral with lower thresholds $\lambda_i - \lambda_j$. Differentiating one threshold gives the negative face integral; the coarea formula for the scalar coordinate $(v_i - v_j) \cdot X$ converts that integral to $-A_{ij}/\ell_{ij} = -w_{ij}$. Thus

$$\partial_{\lambda_i} \gamma_n(C_i) = - \sum_{j \neq i} w_{ij}, \quad \partial_{\lambda_j} \gamma_n(C_i) = w_{ij} \quad (j \neq i). \quad (8b)$$

Since the objective's price gradient is $1/k - \gamma_n(C_i)$, its price Hessian is exactly L . Every weight is positive at every finite price. For any nonzero u with $\sum_i u_i = 0$,

$$u^T Lu = \sum_{i < j} w_{ij} (u_i - u_j)^2 > 0.$$

This gives strict convexity on the price gauge along every nonconstant segment, hence uniqueness of its minimizing price. The gauge gradient maps $\mathbf{1}^\perp$ to itself, and its Jacobian L is invertible there. The implicit function theorem makes balancing prices smooth in the full-rank scores. A smooth positive matrix square-root factorization of $Q|_{\mathbf{1}^\perp}$ then gives smoothness in Q .

For the covariance derivative, set $N = (M^T M)^{-1} M^T$. Since M is a column isomorphism onto $\mathbf{1}^\perp$, $NM = I_n$ and $MN = P$. For the given D let $A = NDN^T$. Then $A = A^T$ and $MAM^T = PDP = D$. For sufficiently small positive and negative t , $I + tA$ is positive, and

$$M(t) = M(I + tA)^{1/2}, \quad M(t)M(t)^T = Q + tD, \quad M'(0) = MA/2.$$

Off Gaussian-null ties the envelope score derivative is the winning moment matrix B ; the difference quotients have an integrable bound proportional to $1 + |X|$. The price derivative vanishes at balance. Using (7) and cyclicity of trace gives

$$d\mathcal{C}(Q)[D] = \frac{1}{2} \operatorname{tr}(B^T MA) = \frac{1}{2} \operatorname{tr}(LMAM^T) = \frac{1}{2} \operatorname{tr}(LD).$$

Here L uses weights A_{ij}/ℓ_{ij} ; it must not be confused with the interface-area Laplacian with weights A_{ij} used elsewhere in multi-bubble profile theory. $\square$

3. The imported perimeter theorem and the regular constant

We use the established Gaussian multi-bubble theorem [MN22, Theorem 1.1]: for $2 \leq k \leq n + 1$ and fixed positive masses in $\mathbb{R}^n$, a simplicial Voronoi cluster of k equidistant sites minimizes the total Gaussian perimeter $S = \frac{1}{2} \sum_i P_\gamma(C_i) = \sum_{i < j} A_{ij}$. We always apply it in dimension $n = k - 1$, with $k = n + 1$ and the strictly positive mass vector $(1/k, \dots, 1/k)$. To match the theorem's cluster definition, take the strict winning cells in (4), or assign ties to the smallest label. These Borel cells are disjoint and cover almost everywhere. Their Gaussian-null difference from (4) changes no mass, moment or variational perimeter. Each has finitely many flat facets of finite Gaussian area, and intersections of codimension at least two contribute no interface area; for $n = 1$ the interface is a point with its zero-dimensional Hausdorff measure. They therefore have finite Gaussian perimeter. The surface weight is the ambient density $\varphi_n = (2\pi)^{-n/2} e^{-|x|^2/2}$ against $\mathcal{H}^{n-1}$, not an independently normalized $(n - 1)$ -dimensional Gaussian density. The original measurable or fractional partition need not have finite perimeter: only this auxiliary polyhedral cluster is used in the imported theorem.

At equal masses this minimizer is the central regular simplex fan: balancing prices of regular scores are unique modulo constants by Lemma 2, and permutation symmetry makes all of them equal. Only the perimeter minimum, not its uniqueness part, is needed below.

For the trace-one regular scores with Gram matrix $Q_* = P/n$, we can write $Y_i = (Z_i - \bar{Z})/\sqrt{n}$. Their winning labels are uniform, so

$$\mathcal{C}(Q_*) = c_k.$$

All facet weights are equal by symmetry; (7) gives $L_* = c_k P$ and $w_{ij}^* = c_k/k$. All pair distances are $\ell_* = \sqrt{2/n}$. Consequently the minimum perimeter is

$$S_* = \frac{k(k-1)}{2} \frac{c_k}{k} \sqrt{2/n} = c_k \sqrt{n/2}. \quad (9)$$

For any $Q \in \mathcal{K}$ positive definite on $\mathbf{1}^\perp$, weighted Cauchy–Schwarz and (7) give

$$S^2 = \left(\sum_{i < j} w_{ij} \ell_{ij} \right)^2 \leq \left(\sum_{i < j} w_{ij} \right) \left(\sum_{i < j} w_{ij} \ell_{ij}^2 \right) = \frac{1}{2} \operatorname{tr} L \mathcal{C}(Q).$$

Combining this with $S \geq S_*$ proves the fundamental inequality

$$\mathcal{C}(Q) \frac{\operatorname{tr} L}{n} \geq c_k^2. \quad (10)$$

All weights here are positive, so equality in weighted Cauchy is possible only when all pair distances are equal. This observation will supply the equality classification.

4. The radial differential comparison and its deficit

Theorem 3 (covariance form). For every $Q \in \mathcal{K}$, $\mathcal{C}(Q) \leq c_k$, with equality only at Q_* .

Proof. Put $Q_t = Q_* + t(Q - Q_*)$. It is positive on $\mathbf{1}^\perp$ for $0 \leq t < 1$, even if the endpoint Q is singular. Let $C(t) = \mathcal{C}(Q_t)$ and L_t be its facet matrix. For $0 < t < 1$, (7)–(8) give

$$tC'(t) = \frac{1}{2} \operatorname{tr}(L_t(Q_t - Q_*)) = \frac{1}{2} \left(C(t) - \frac{\operatorname{tr} L_t}{n} \right). \quad (11)$$

Let $h(t) = C(t)^2 - c_k^2$. Inequality (10) implies

$$th'(t) = C(t)^2 - C(t) \frac{\operatorname{tr} L_t}{n} \leq h(t), \quad \left(\frac{h(t)}{t} \right)' \leq 0. \quad (12)$$

At zero $L_* = c_k P$, and $\operatorname{tr}(Q - Q_*) = 0$. Thus (8) gives $C'(0) = 0$, while $C(0) = c_k$. It follows that $\lim_{t \downarrow 0} h(t)/t = 0$. The quotient is nonincreasing by (12), hence $h(t) \leq 0$ on $(0, 1)$. Continuity (5) gives the claimed bound at $t = 1$. No differentiability, balancing-price limit or bounded limiting facet matrix at a singular endpoint is required.

If $\mathcal{C}(Q) = c_k$, the quotient tends to zero also at one. A nonincreasing function with both endpoint limits zero is identically zero. Therefore $C(t) = c_k$ for every $0 < t < 1$, and (11) gives $\operatorname{tr} L_t = nc_k$. The perimeter/Cauchy chain in Section 3 then consists entirely of equalities. All $\ell_{ij}(t)$ are equal, since all weights are positive. Centered scores of squared norm sum one with equal pair distances have Gram matrix P/n : summing their squared pair distances over j gives $k|v_i|^2 + 1$, independent of i ; thus $|v_i|^2 = 1/k$ and $\ell_{ij}^2 = 2/n$. Hence $Q_t = Q_*$. For any fixed $0 < t < 1$, this forces $Q = Q_*$. $\square$

There is also an exact deficit identity. For Q_t as above put

$$D(t) = C(t) \frac{\operatorname{tr} L_t}{n} - c_k^2 \geq 0.$$

The equality version of (12) reads $(h/t)' = -D(t)/t^2$, giving

$$c_k^2 - \mathcal{C}(Q)^2 = \int_0^1 \frac{D(t)}{t^2} dt. \quad (13)$$

This is a nonnegative improper integral. At zero smoothness gives $h = O(t^2)$; integrating on compact subintervals and using the endpoint limits proves convergence and (13), even for a singular endpoint. In particular, writing $W(t) = \sum_{i < j} w_{ij}(t)$,

$$D(t) = \frac{2}{n} \left[W(t) \sum_{i < j} w_{ij}(t) \ell_{ij}(t)^2 - S(t)^2 + S(t)^2 - S_*^2 \right]. \quad (14)$$

Both terms are nonnegative: the first is the weighted edge-length variance, and the second is the established perimeter deficit. This is an exact identity, not a claim of an explicit metric stability constant or of full covariance concavity.

5. Return to measurable and fractional partitions

For any partition in Theorem 1, let M be its $k \times d$ moment matrix. Its rows sum to zero, so MM^T has rank at most $k - 1$ and trace F . For every price, its labels give a feasible assignment in (3):

$$F = \mathbb{E}[m_I \cdot X] \leq \mathcal{C}(MM^T).$$

This remains valid for fractional labels by replacing the selected score with their convex combination. The Gaussian score distribution has the same covariance even if the original dimension differs from $k - 1$. When $F > 0$, (5) and Theorem 3 give $F \leq c_k \sqrt{F}$, proving (2). For $F = 0$ the positive bound is strict.

Equality forces $MM^T = F_k P/n$ and equality in the feasible assignment. The regular centered scores have null ties and equal prices by symmetry and uniqueness, so the original labels must be their winning labels almost everywhere. Fractional labels likewise must be the unique winning indicators. Their span has dimension $k - 1$, which excludes equality if $d < k - 1$.

Conversely, take the regular scores with Gram matrix P/n and their central winning cells. By (7) and $L_* = c_k P$, their actual moments are $c_k v_i$, of squared sum c_k^2 . Cylindrical extension preserves both masses and moments. This proves attainment and all equality assertions. For $k = 2$ the trace-one covariance domain is a singleton; the proof includes this case. The excluded $k = 1$ case is separately trivial, with one cell and zero first moment.

6. Mathematical and verification dependencies

The only imported non-elementary geometric result is the Gaussian multi-bubble perimeter theorem [MN22]. Its hypotheses, ambient dimension and perimeter normalization are stated in Section 3. The proof uses Gaussian integration by parts, dominated convergence, smooth matrix square roots in the positive cone, the implicit function theorem, and one-variable differential comparison. Price derivatives are justified by Gaussian differentiation in fixed coordinates in Lemma 2. No further regularity or second-variation theory for arbitrary perimeter minimizers is required beyond the

imported theorem. No local simplex Hessian, full covariance concavity or mountain-pass theorem is assumed.

No finite search, solver, floating integral or numerical eigenvalue is a premise of this result. Partial Lean proofs and exact arithmetic controls bundled in this paper package have their own explicit scope and do not formalize its Gaussian endpoint. A separate Lean development formalizes actual Gaussian moments, balancing prices, flux and the comparison and equality chains, with the needed geometric perimeter lower bound remaining an explicit premise. It also proves regular attainment and the necessary simplicial BV upper bridge. This does not give a complete Lean proof of the imported multi-bubble theorem or the unconditional all- k endpoint. Internal model reviews are not external human peer review. The complete earlier four-cell proof [FOUR] is preserved as a historical, longer argument; the present proof is standalone and covers all k .

References

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